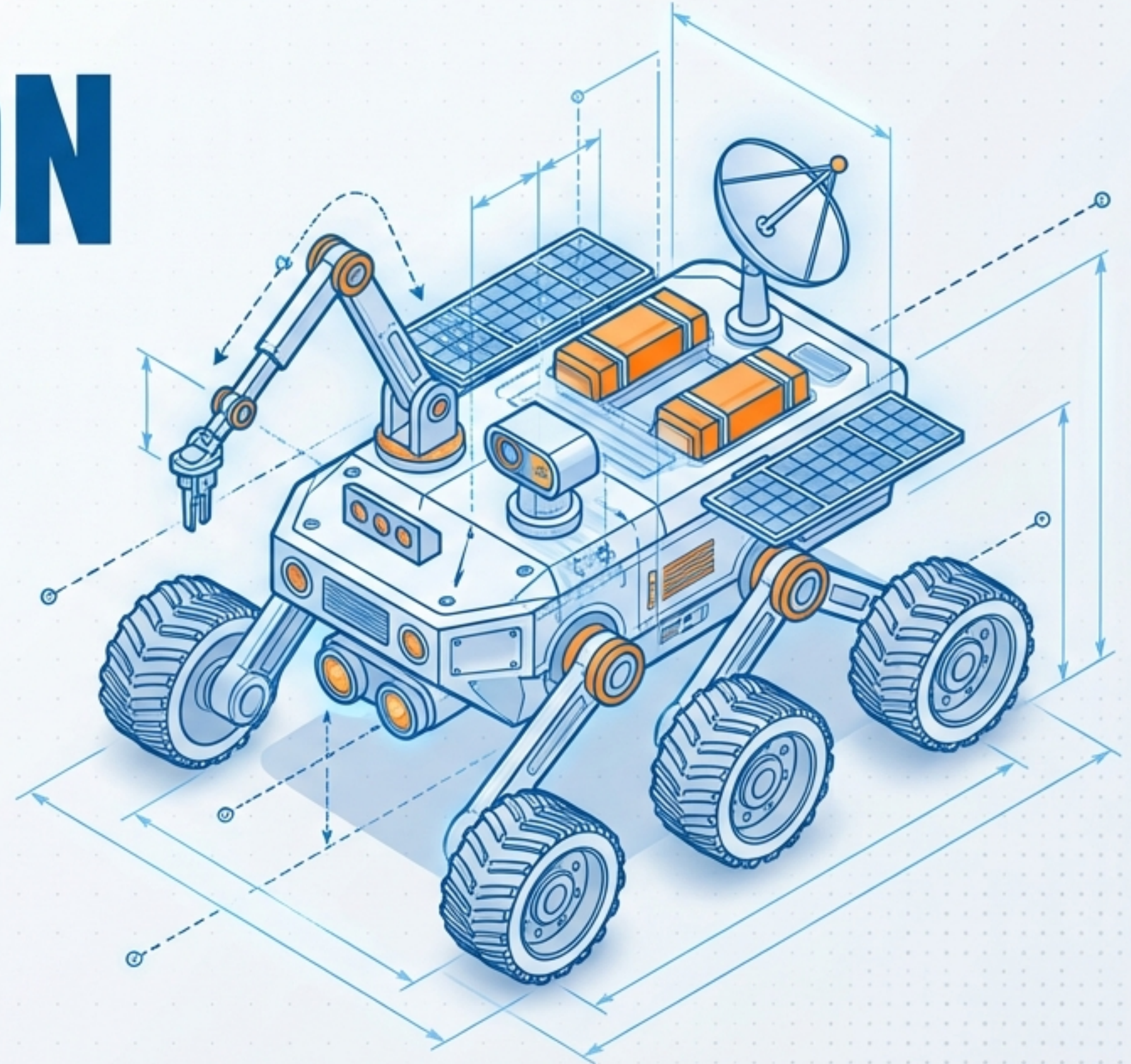


INTRODUCTION TO ROBOTICS

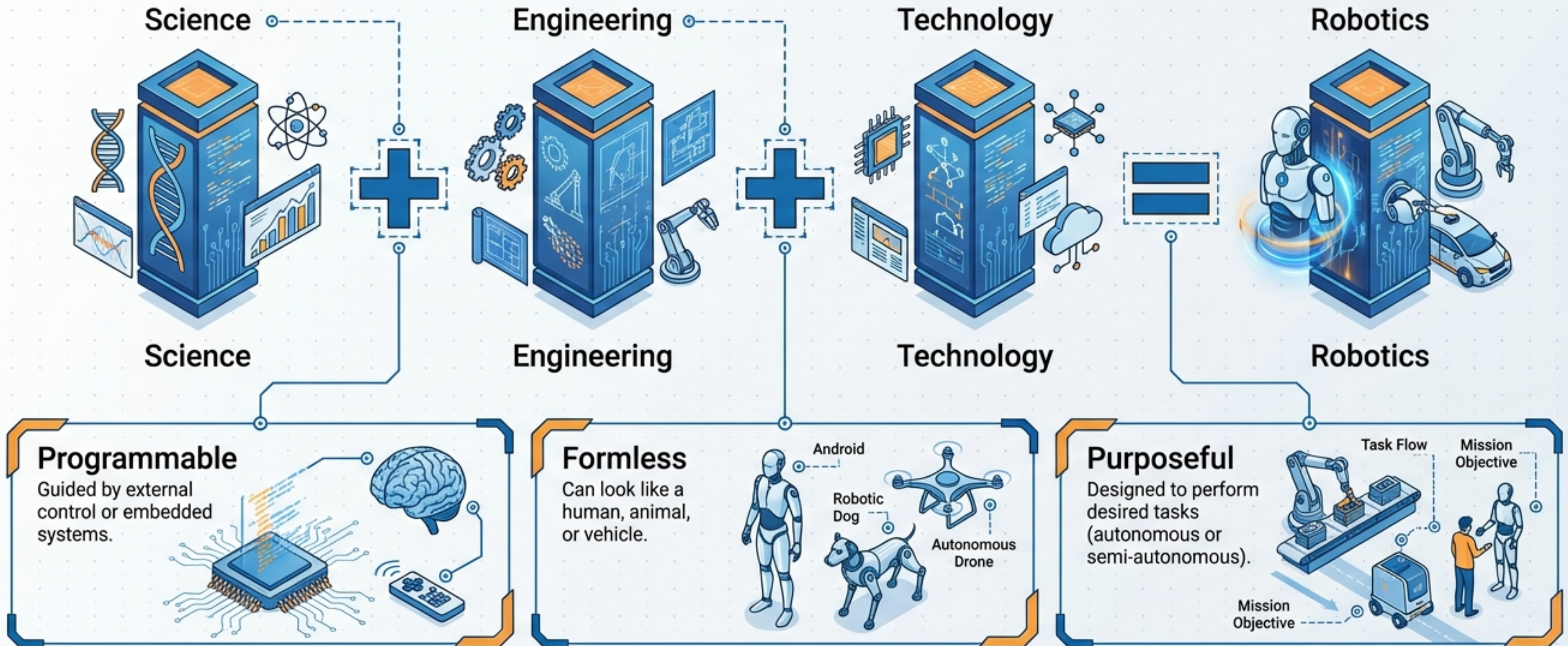
The Future Innovator's
Blueprint

Junior Engineering Module | Level 1

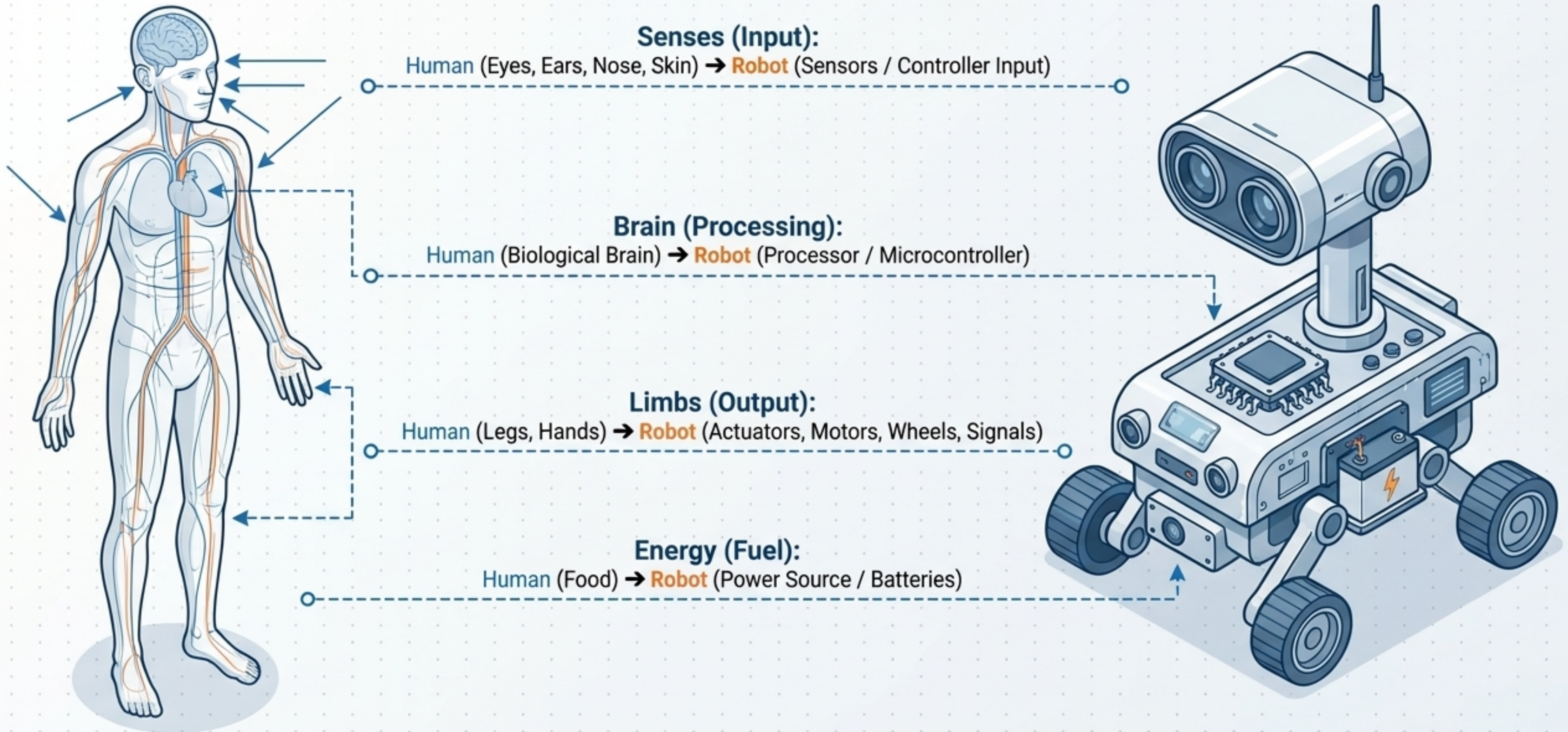


What Makes a Machine a Robot?

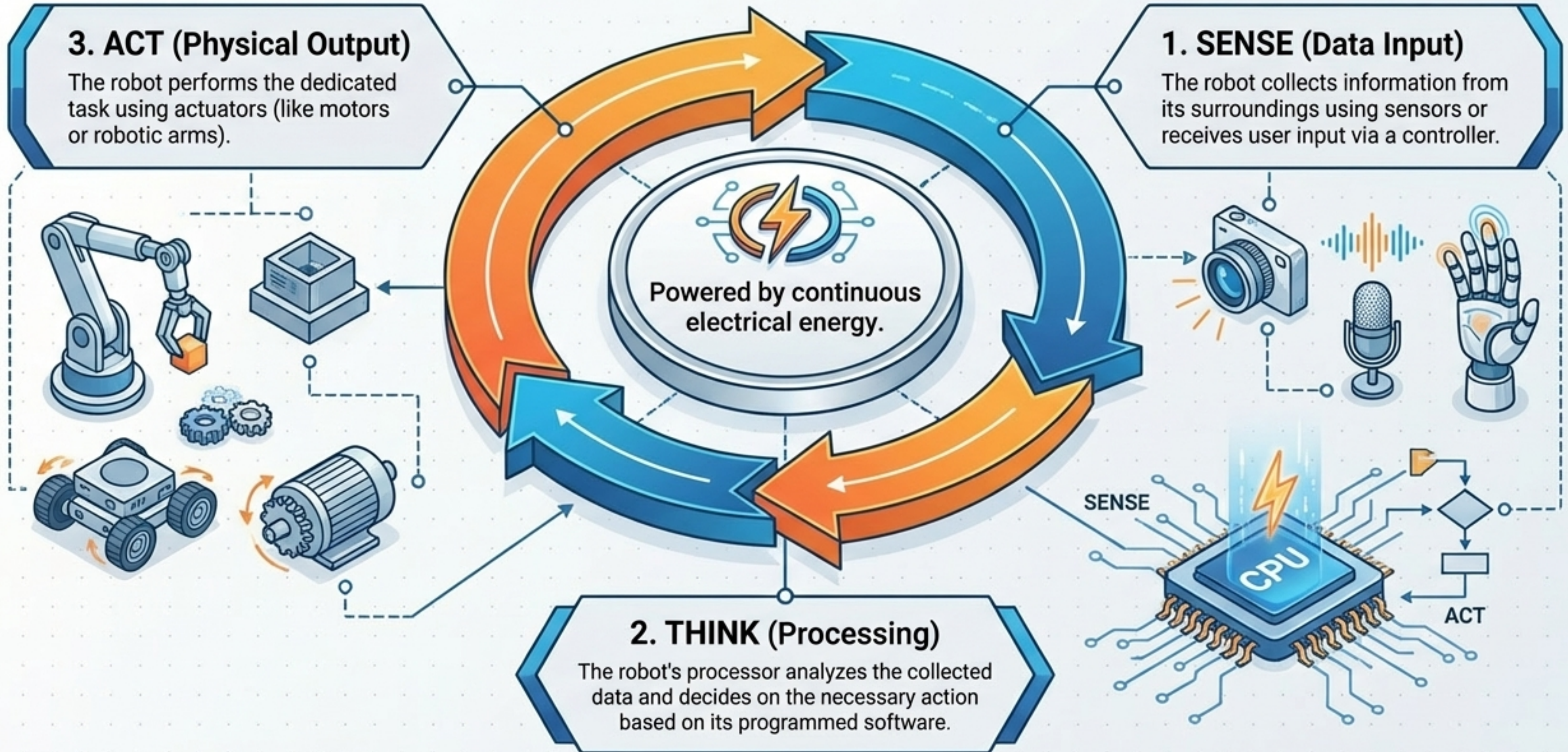
A **robot** is a machine programmed by a **computer**, capable of automatically carrying out a complex series of actions.



Anatomy of a Robot: The Biological Mirror



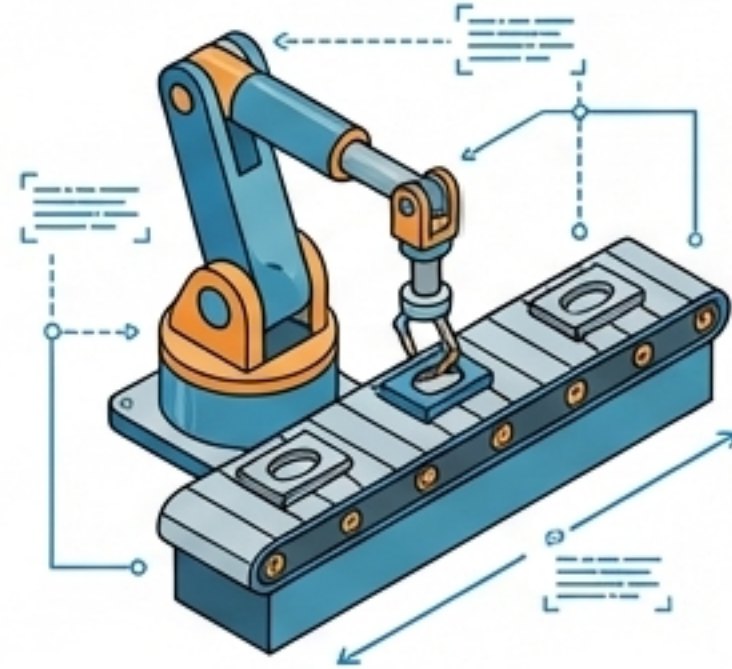
The Core Framework: Sense, Think, Act



Robots in the Real World

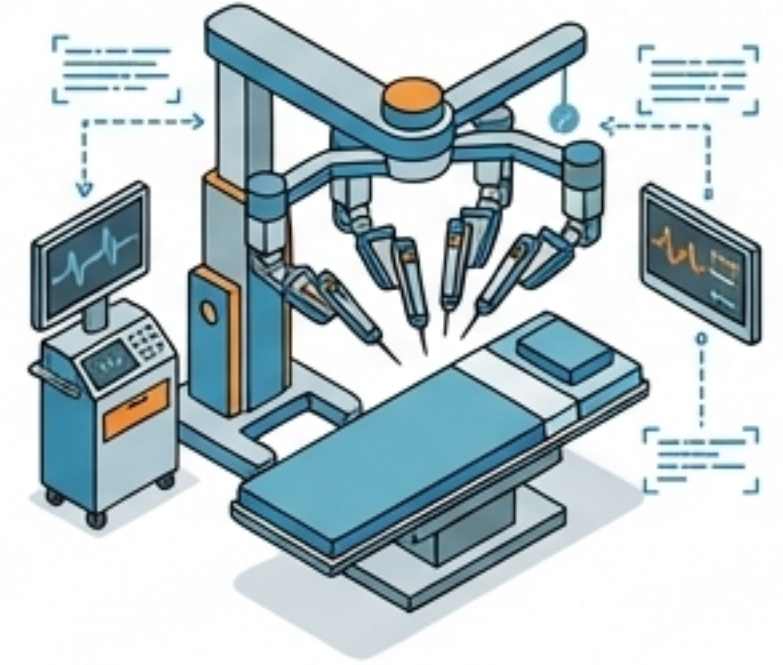
Manufacturing Industries

Handling materials, processing operations, assembly, and precise inspection.



Medical Industry

Assisting doctors in complex surgeries, lab automation, and medical device packaging.



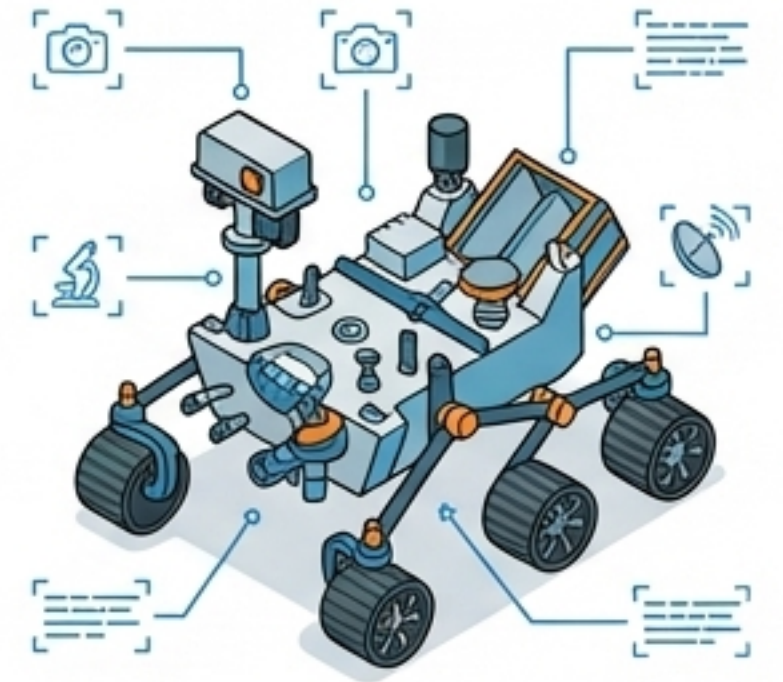
Military

Tending to injured personnel and operating in hazardous zones.



Research Fields

Conducting exploration in unknown territories, underwater depths, and other planets.



Zooming In: The Robotic Vehicle

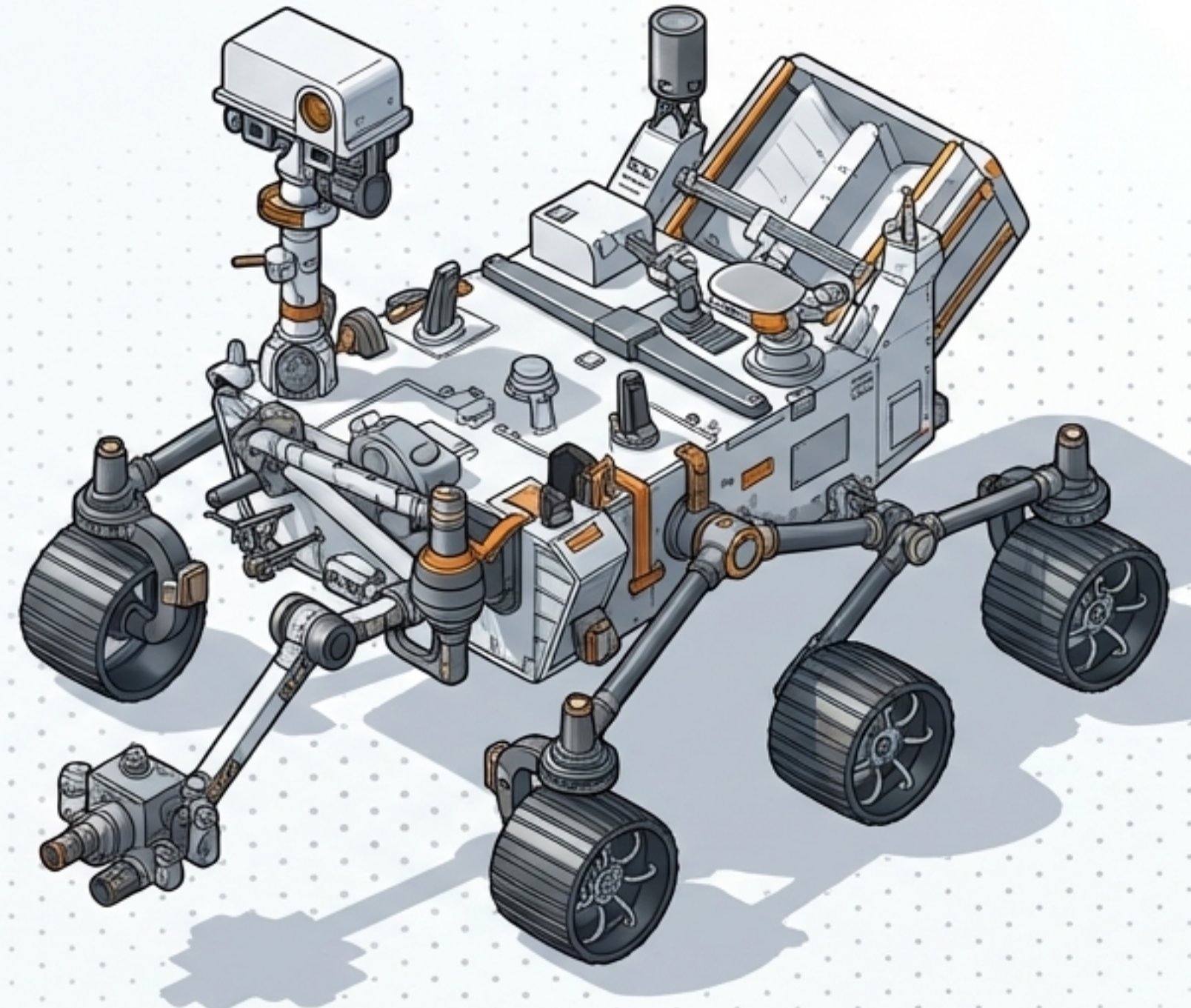
The most common robotics application.

Definition

A special type of machine designed to move on the ground, in the air, under water, and in space without a human on board.

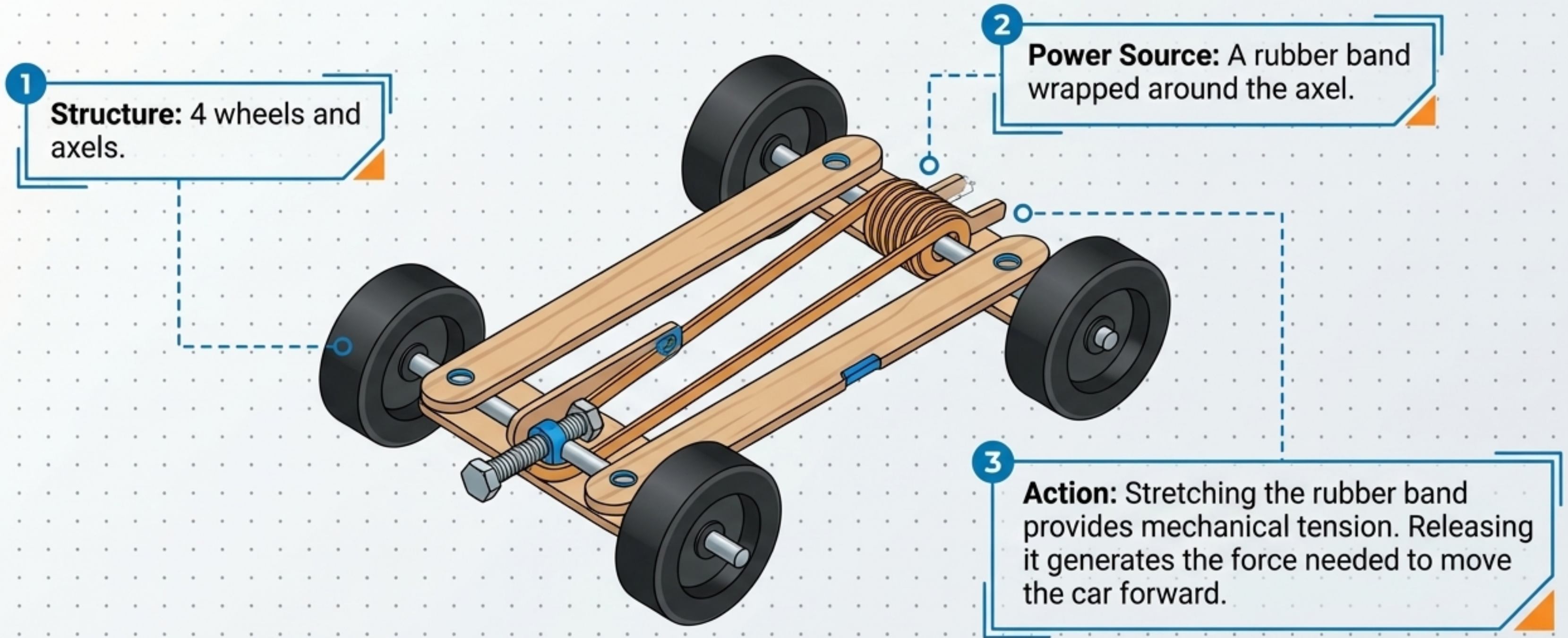
Key Characteristic

Controlled and operated by sensor-based control systems rather than manual driving.



Evolution of Control: Uncontrolled Mechanisms

A basic robotic vehicle built without electronic devices. It has a limited distance and cannot be steered once released.



Evolution of Control: The Base Bot

A simple, semi-autonomous robot that handles tasks based entirely on user inputs.

The Foundation

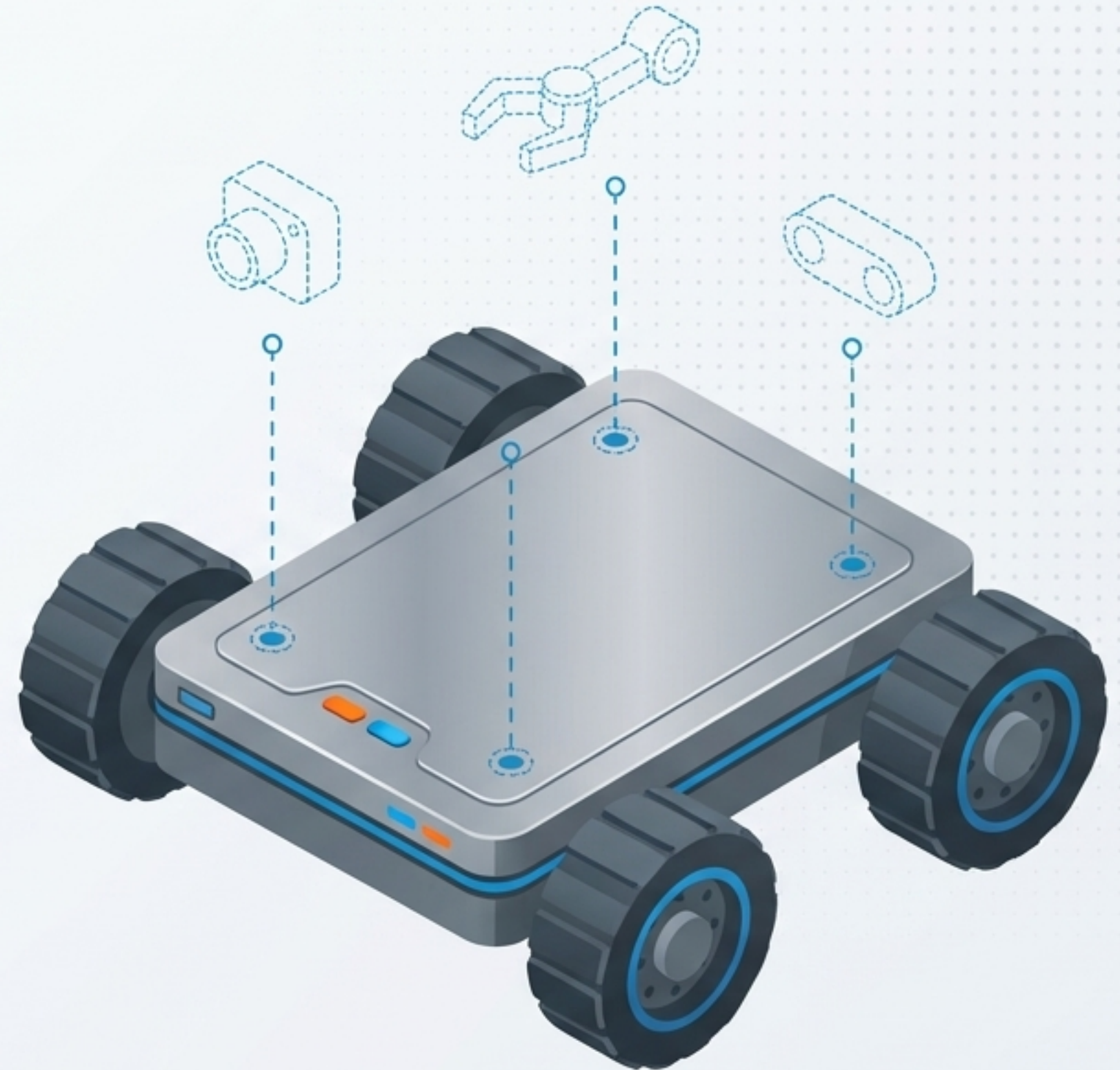
It is the starting point. You must build a solid base before adding complex features.

User Controlled

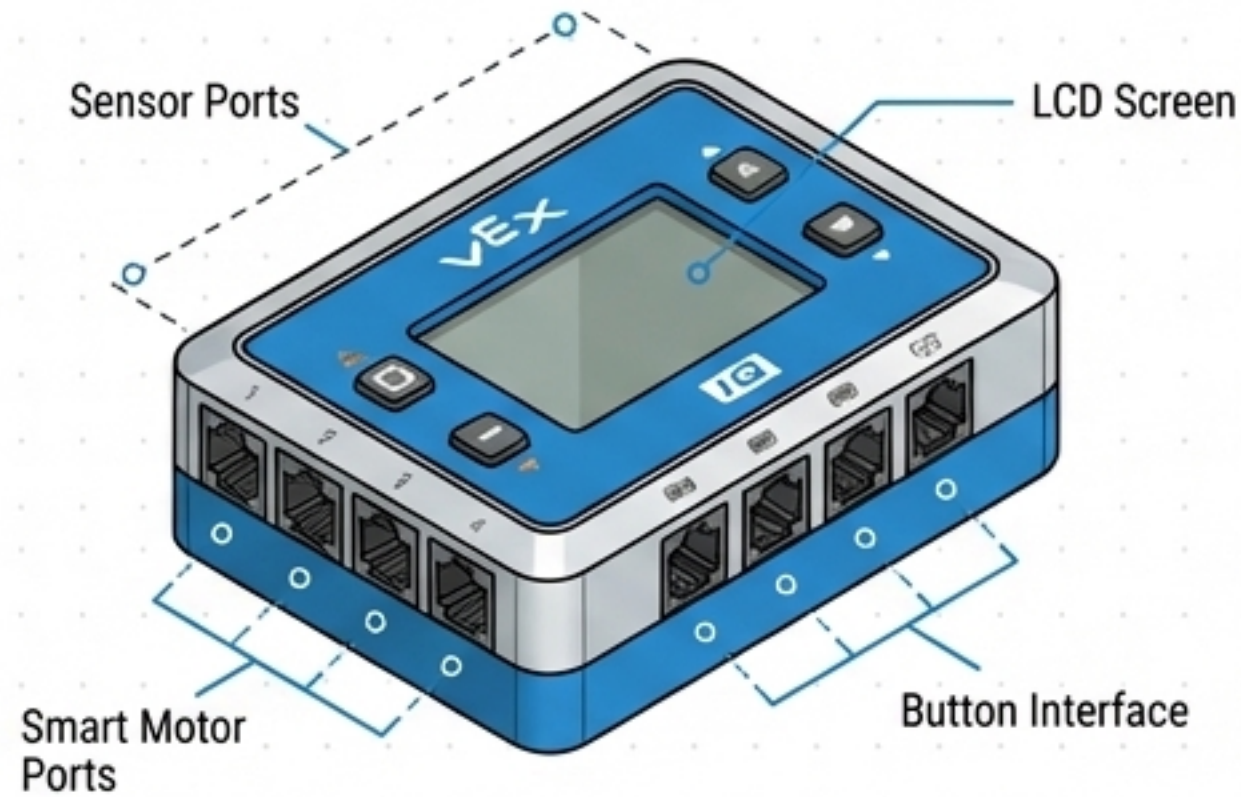
Movement is directed by a human using a wired or wireless controller.

Expandable

Designed to be modified into complex bots by adding multiple sensors for different applications.



Anatomy of a Base Bot: Hardware



The Brain (Processor)

- The processing unit of the robot.
- Gets input from sensors and makes decisions.
- Contains multiple ports to connect sensors and actuators.

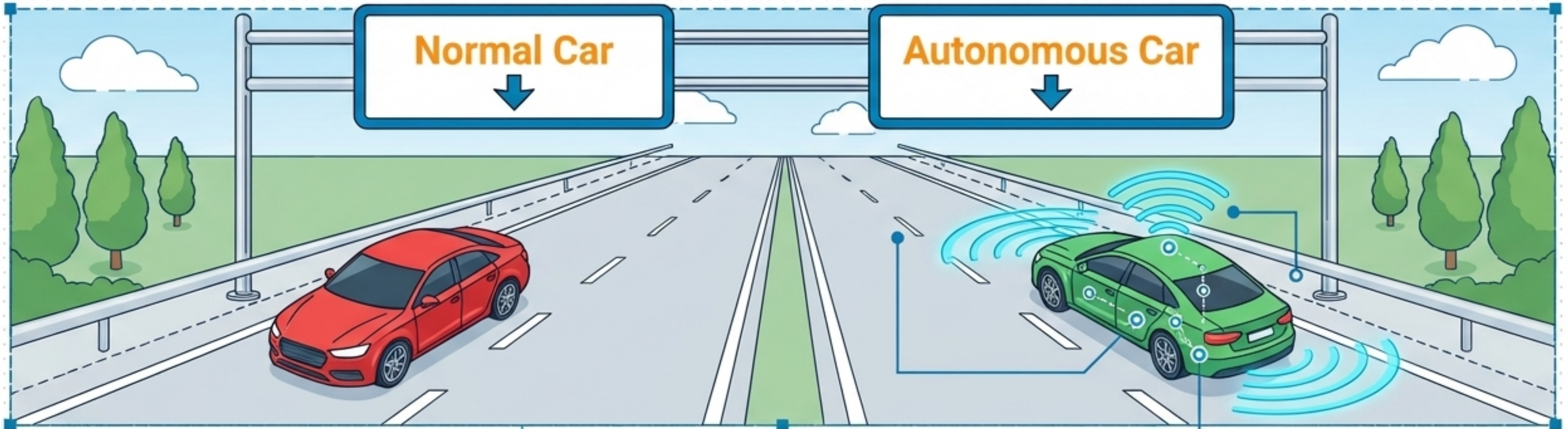


The Controller (Joystick)

- Provides wired or wireless user input.
- Features two analogue joysticks for fine driving/turning control.
- Features eight buttons for controlling additional mechanisms (like raising an arm).

Evolution of Control: Fully Autonomous

When hardware is paired with advanced software (programs), the robot can operate entirely without human intervention.



Normal Car (Semi-Autonomous/Manual):

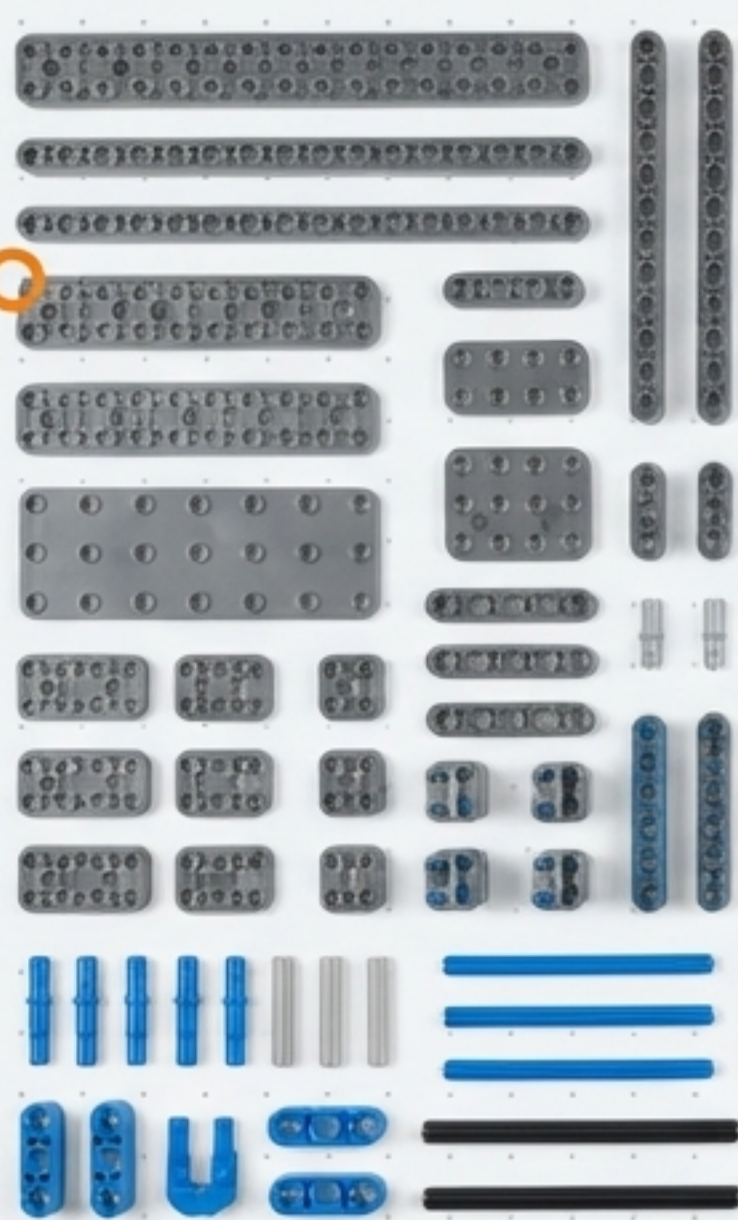
Requires continuous input from a human driver.

Automated Vehicle (Autonomous):

Gathers information from surroundings via sensors, processes it, and steers itself. No human input required.

Inside the Robotics Toolkit

Structural Beams
& Plates



Gears & Pulleys



Tires & Hubs



Smart Motors
(Actuators)



The Brain
(Processor)



The Controller &
Tether Cables



Activity: Building the Base Bot

1

Gather & Sort

Collect all required parts from the toolkit to ensure a smooth build.



2

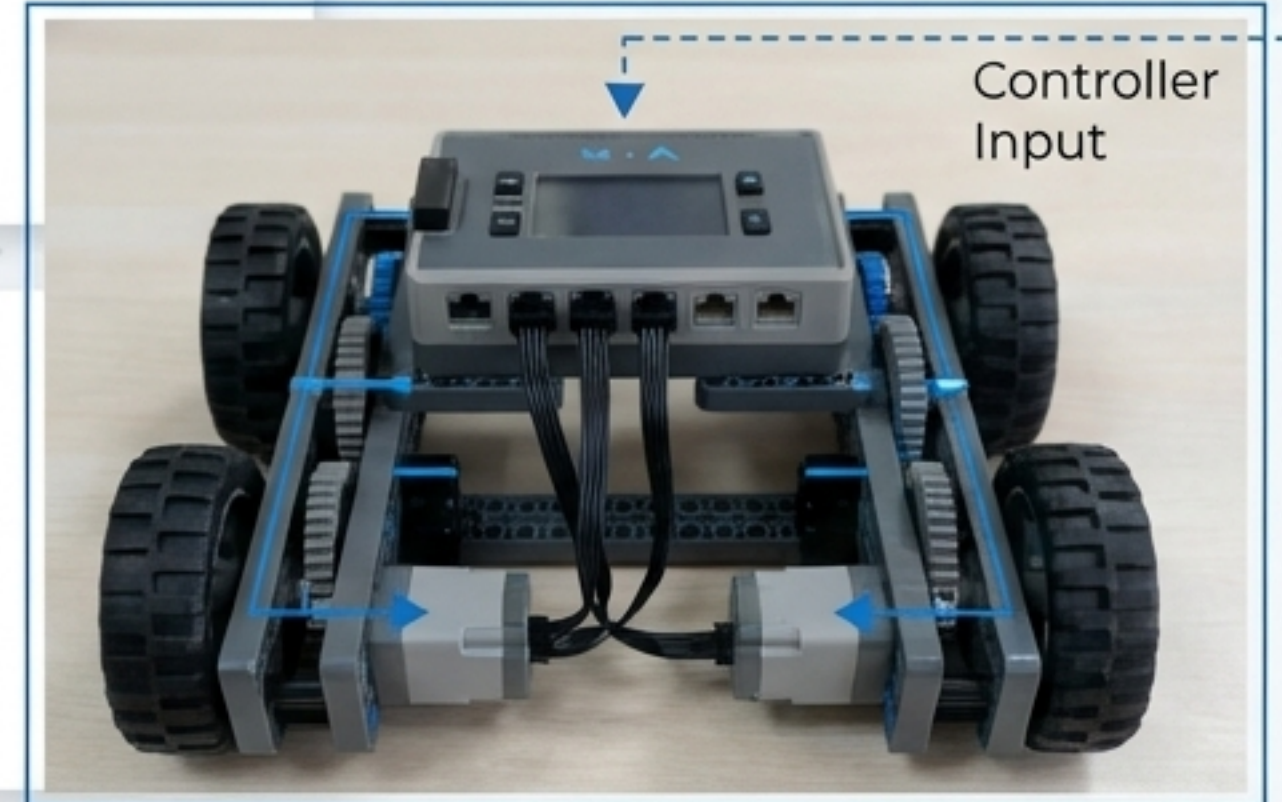
Build the Drivetrain

Construct the structure connecting the smart motors to the wheel-gear assemblies to provide movement.

3

Wire the Brain

Connect the motors and wheel-gear structures to the Brain (processor) using connecting wires. Finally, connect the Controller to provide user input.



Robotics Lab Safety Protocols



Rule 1: Eye Protection

Always use clean protective glasses when working with flying objects or projectiles (loose parts).



Rule 2: Supervision

Always perform experiments with approval and under direct adult/teacher supervision.



Rule 3: Clear Floors

Never leave robotics pieces on the floor to prevent tripping hazards.



Rule 4: Electrical Safety

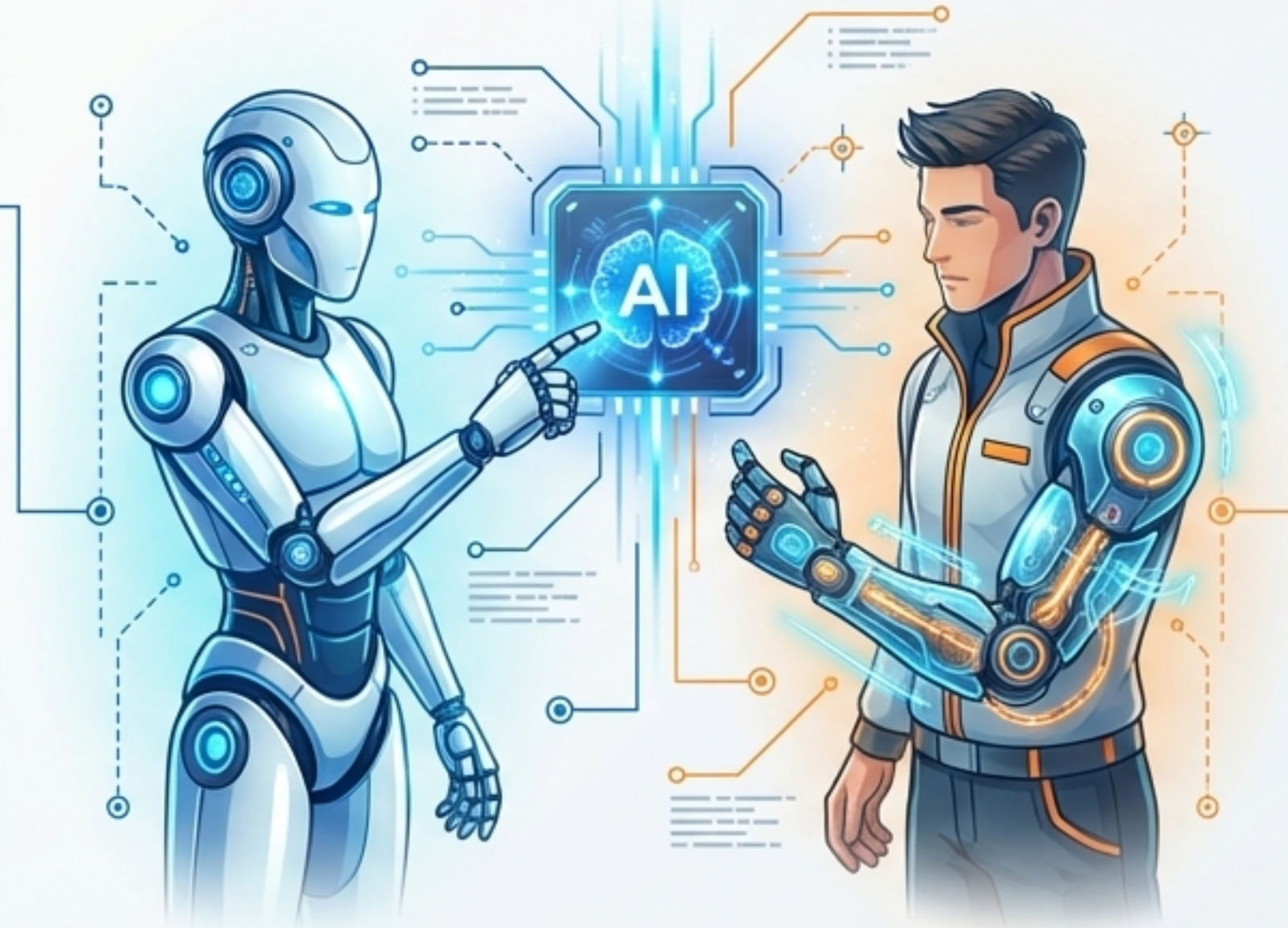
Do not work with robots around water or with wet hands—robots and computers run on electricity!

The Future: Beyond Basic Programming

Artificial Intelligence (AI)



The capabilities of robots are expanding exponentially by implementing AI, allowing them to learn and adapt rather than just follow rigid code.



Human Augmentation



Technology that enhances human capabilities or replaces lost ones. This can artificially increase physical strength, enhance sight, or boost human intelligence.

System Diagnostic: Knowledge Check

Fill in the Blanks

1. Robotics integrates science, engineering, and technology to develop machines called robots.
2. A robot senses information using inputs from sensors/controller, analyses using a processor, and acts using actuators/motors as an output.

True or False?

1. Sensors decide the action to perform based on user input.
2. An automated vehicle does not require any input from the user/driver.

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